



*The Epistemological Positioning of Technological Determinism Theory
in the Arab Media Studies in the New Communicative Environment:
Critical Analytical Study on a Set of New Media Studies in Algeria*

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Abstract ;

The ideas of Marshall McLuhan are a prominent reference for information and communication scholars. However, in Algeria, they raise a hot epistemological debate, mainly with the current digital transformations. In this regard, this paper aims at reviewing his approach to the media environment to interpret the current variables in the digital world. In so doing, we conduct a critical analysis of the main Algerian works that referred to the theory in tackling the new media to reveal the patterns of use and focus on the theoretical and methodological introductions. Findings show that these ideas are seen as a historical heritage, a criticism topic, and an outlook source.

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1. Introduction

The epistemological positioning of communication theories is one of the main issues in the light of the digital transformations, as many studies found out that traditional theories need revision to renew their theoretical references and interpret the complicated changes in the new communication environment. In this context, McLuhan's approach has influenced the media studies since the 1960s thanks to the different perception of the relation between the medium and the society. The approach states that media are not mere tools to convey messages; rather, they are factors that can reshape the human perception and restructure the socio-cultural tents without any resistance.

With the new media environment that witnesses the accumulation of technological media and tools, scholars refocused on this theory under the new media studies, questioning its interpretational abilities and capacity to cover the current transformation, mainly with the emergence of modern cognitive paradigms that described his ideas with determinism. This theory was incarnated in academic papers; however, its presence was not unified, as some scholars described its quotes, others used it critically by linking it to the local digital transformations, while others refused it, claiming it is not sufficient to understand

the complexities of the current communicative reality.

Based on what was said, this paper raises an epistemological problematic that states, "to what extent is technological determinism theory a valid epistemological frame to interpret communicative variables in the Algerian media studies? How does this theory interact with the modern media paradigms? What position does it hold in the Algerian media studies?". In answering these questions, we use a critical analytical approach based on a theoretical and methodological review of the main Algerian media studies that focused on the theory in order to find out the descriptive and critical perspectives and predict the possibility of developing a cognitive perspective that suits the Algerian communicative environment.

Hypotheses:

Based on literature review and the ideas of McLuhan, we hypothesize that:

- The presence of the technological determinism theory in the Algerian media studies takes different aspects, including descriptive adoption, critical use, and refusal; what reflects the lack of a unified epistemological view.
- The technological determinism theory is limited and cannot be

adapted to the Algerian local context.

- Literature review shows that the technological determinism is not sufficient to understand the complexities of the new communication environment; therefore, it is necessary to rely on additional cognitive paradigms.
- Repositioning the technological determinism theory under the modern paradigms is a step to enrich the Algerian media studies and foster their ability to interpret and analyze the current digital challenges.
- Technological determinism is a theoretical introduction to study the electronic media.

Importance of the study:

The importance lies within the topic and problematic that tackle the positioning of the technological determinism in the Algerian media studies as one of the main theories that investigate the relation between technology and society to reveal the limits of the prevailing descriptive use, show the need for deeper critical studies that meet the requirements of the new communication theory, and assess the Algerian studies in this field to allow for

establishing approaches that suit the local status-quo.

Methodology:

This paper is part of the critical analytical studies that analyze the elements of the phenomenon, and provides an objective criticism based on some comparative theories, deduction, and reasoning.

Population and Sample:

The population covers all the topics that tackled, directly referred to, or described technological determinism. The sample includes 06 papers chosen on purpose. From the Algerian Scientific Journals Platform ASJP.

Study Axes:

1. The cognitive background of technological determinism: genesis and pillars.
2. The presence of technological determinism in the new media studies in Algeria: discussion and analysis.
3. Towards the epistemological repositioning of the theory in the Algerian media studies under the digital environment.

The pillars of McLuhan Thoughts in Media Studies:

2.1 Before discussing the history and intellectual tent of the theory, we have to

tackle some concepts, such as the concept of theory.

Theory: Based on some definitions, we can conclude that it is the facts and observation-based assumptions that aim at explaining a given phenomenon. It can find and explore new methods and questions for a continuous research. This applies to McLuhan's theory.

Media studies: They are the theoretical or empirical studies that handle new media topics that link between the social phenomena that appeared after the appearance of internet.

The theory of Marshall McLuhan: It was established by Marshall McLuhan 1911-1980 (Balle, 1972) in 1964 as part of the theories of media outlets' strong effect. The theory states that the influence of technology is not limited to ideas and concepts; rather, it changes the relations between senses and sensor perception models without any resistance (Al Ayadi, 2018). In addition, the theory confirms that the technological media basically affect societies, and that the development of communication tools and processes is tightly related to the life and human civilizations (al Mazahra, 2014). McLuhan inspired his ideas from Adam Smith, John Stuart Mill, and Nietzsche; therefore, he attributes any social change to the material

power and the control of machines. Besides, he embraced the ideas of Harolld Agnes regarding the bias of communication, which he described with the "global village", predicting the emergence of a small global village that has no spatial or temporal limits. The works of McLuhan helped found modern information sciences, as he tackled the relation between technology, culture, and human communication. The following are some of his works:

- The Mechanical Bride: Folklore of Industrial Man (1951)
- Explorations in Communication (1960) – with Edmund S. Carpenter
- The Gutenberg Galaxy: The Making of Typographic Man (1962)
- Understanding Media: The Extensions of Man (1964)
- The Medium Is the Message: An Inventory of Effects (1967) – with Quentin Fiore
- Verbi-Voco-Visual Explorations (1967)
- War and Peace in the Global Village (1968)
- Through the Vanishing Point: Space in Poetry and Painting (1968) – with Harley Parker
- Counterblast (1969)

- The Interior Landscape: The Literary Criticism of Marshall McLuhan 1943–1962 –edited by Eugene McNamara
- Culture Is Our Business (1970)
- From Cliché to Archetype (1970) – with Wilfred Watson
- Take Today: The Executive as Dropout (1972) – with Barrington Nevitt
- City as Classroom: Understanding Language and Media (1977) – with Kathryn Hutchon & Eric McLuhan
- Laws of Media: The New Science – with Eric McLuhan (published after his death)

2.2 The Intellectual Pillars of McLuhan's Theory:

Technological determinism is based on three main assumptions:

First: Media outlets represent the sensor perceptions and cognitive rules of people inside the society.

Second: Media outlets redistribute the cognitive patterns on all the social patterns.

Third: Every media outlet restructures the symbolic and cultural environment. This is confirmed by his quote, “We make our tools, but our tools shape us”. This shows the structuralist nature of the technological variables in culture, as the technological revolutions sometimes lead to deep social

changes in societies; such changes cannot be resisted without knowing how they had happened.

The first assumption pushed McLuhan to believe that numbers, games, and money are extensions of the human senses, as everybody lives within a given communicative environment that directs his perception and existentialist experience. In this context, McLuhan sees that games are a tool for learning and socialization that allows the individual to experience his identity within symbolic interactions. In addition, money expand the society's ability to exchange values and meanings and represent an extension of the human sense. Thus, we can say that the technological developments continuously reshape the social structure through transformations in the patterns of communication and knowledge. Besides, McLuhan shows that language, writing, money, and games are human extensions that reshape and direct the human experience.

Currently, social media have the same roles of traditional outlets, such as cinema, radio, and TV, as they turn separate individual practices into a wide-range collective activity and create new common patterns of social interactions. Most of social media users adhere to virtual

spaces that have the features of local and tribal communities, making these platforms a reflection of the global village. In this context, they do not only gather people in one communicative environment; rather, they reshape the different dimensions of the human life. Therefore, McLuhan did not provide any given definition to media and sees that they are effective tools in shaping the perceptions of the world. The affective messages allow people to view issues from perspectives of these media, regardless their positive or negative content. Besides, they distribute sensor and cognitive patterns on all the social categories (Azam et al., 2020).

Third: Media outlets turn the world into an integrated connected economic, political, social, and cultural system, as described by McLuhan as a global village (Coupland, 2010). This assumption embodies McLuhan's vision on the integration of humanity within a unique international net. The appearance of social media fostered this view, as the digital connections redefine the geographical limits and allow for immediate transfrontal interactions.

- Medium, time, place: This principle is the outcome of Hallond Agnes influence on McLuhan. He used his assumption on space and time to interpret the role of communication media in the continuity of

societies. McLuhan confirms that any hot or cold medium has features of heaviness/lightness and eternity/transience, which allow it to control place or time (Al Ayadi, 2021). This principle is vital and was the starting point for theoretical approaches "the theory of the medium" of in his "No Joshua Meyrowitz Meaning for the Place in the Age of Electronic Media" (al Ayadi, 2021).

- The societies' view to media outlets determines their penetrability and influence because if they are seen as tools for information, entertainment, and learning, people would show more interest in their contents and use far from the obligations of technological determinism. However, if seen as part of the technological development, we face the obligations of the technological determinism.

McLuhan divided the human societies' development into 04 phases based on the prevailing communication patterns, as follows:

- The prior phase where the collective oral communication had prevailed.
- The writing phase that marked shift towards writing thought.
- The printing phase from 1500 to 1900 (Al Mazahra, 2014), which

allowed for spreading knowledge and promoting individualism.

- The electronic phase that marked the return to the oral communication, where electronic information and communication technologies dominated the scene and allowed for a transfrontal immediate communication.

The Presence of Technological determinism Theory in New Media Studies in Algeria: Analysis and Discussion:

2.1 The Positioning of Theory in the Algerian Media Studies:

Table 1. The Positioning of Technological Determinism Theory in the Algerian Media Studies in the Digital Present: The Academic papers as a Model:

Title of the paper	Author	Formal description of the paper	Date of publication
Thinking with McLuhan against McLuhanism in the era of Digital Media	Dr. Nasr al Dine Ayadi	The paper is a critical study and includes 47 pages that revolve around the position of the theory in the information and communication sciences. It tries to examine whether its absence reflects the inconsistency in identifying its position or vagueness. In addition, the paper sheds light on the theory in the digital environment and whether it facilitated our understating of theory. Besides, it questions the epistemological value of the theory in the Arab studies. The paper started with the epistemological and methodological positioning and relied on complexity and social construct paradigms in analysis (Al Ayadi, 2021). Element two: Marshall McLuhan and his sources of inspiration. Third: The intellectual pillars of the theory. Fourth: Refocus on the theory. Then, the paper tackled the position of the theory in the Arab studies, the digital McLuhanism, and the criticism of McLuhanism.	2016
Technological determinism: a theoretical introduction	Souiqat Lobna & Abd al Ilah	includes 25 pages and revolves around the technology on the social patterns and values. It is divided into three main elements, mainly the technological determinism from Agnes to McLuhan, the	2016

to study the uses of electronic information	Abd al Kader	the theory units, and the uses of electronic media of the technological determinism (Al Kader,	
McLuhan Marshall: reading in his theories between the past and the present	Nour Dine Touati	includes 14 pages and revolves around the life of ideas of McLuhan, the phases of human information development, and the basics of his theory (13)	2013
Approaching the technological determinism: historical analysis	Rachid Ben Rached & Hassina Belhadj	includes 18 pages and uses the analytical method to examine whether there is a technological determinism for information and communication tools. It was divided into three parts, the study methodology, structure, importance, and concepts. Then, the study discussed the pillars of the theory and conducted a conclusion (Rached, 2019).	2019
The Algerian society and technological determinism: analytical study on how the Algerian society dealt with the new media	Abd el Moumen Bechibech	includes 10 pages and investigates whether technological determinism changed the communicative structure of the Algerian society. The paper is divided into three parts, a conceptual reading of the new media, the traditional Algerian pattern, the uses of new media in the Algerian society and the technological determinism, and conclusion.	2022
The uses of smart phones by the Algerian youths: analytical study based on McLuhan background	Rachid Ben Rached	includes 23 pages and examines whether there is a fatality in the digital environment and the theory assumptions in the student's uses. The study uses the desk method, discusses some concepts, reviews literature to analyze and discuss the theory (Rached, 2022).	2022

Source: By the Authors based on the outputs of ASJP

Based on these papers, we find out that the Algerian scholars, regardless their backgrounds and affiliations, show interest in the theory. In this regard, most agree on the main assumptions and

definitions of the theory. The paper of Nour Dine Touati is a good reference for these studies because he was the first to shed light on the notions of McLuhan and to introduce the idea of technological

determinism. In his book “Understating Media Outlets: Humanity Extensions” he argues that al Ayadi (2021) uses the expression “McLuhan thought” instead of “Determinism” in the end of the paper, and that any determinism is an exclusion of variables that may be vital for the study of the media, and that they face complicated critical effort and provides ready-made answers. Thus, he calls for overcoming cliché in media studies and for avoiding the quantitative methods that provide given answers that lack analysis and discussion.

Some scholars adoption of the most famous assumptions of McLuhan, mainly “the medium is the message”, “the medium is an extension to the human senses”, “the hot and cold media outlets”, etc. for description and interpretation without any criticism makes us say that this adoption is descriptive. In addition, some scholars rely on the quotes and assumptions without questioning, comparison, and criticism despite that all their criticisms were based on the ideas of Richard Black and on other Western scientific thoughts. For instance, a study criticized the theory from the perspective of Azzi Abd al Rahman and Malik Ben Nabi (Bechibech, 2022). This variation in adoption may reflect a lack of a unified epistemological vision towards the theory; this goes with the first hypothesis of the study. Most of the problematic included criticism to the theoretical principles,

assumptions, and results. This reflects the epistemological examination that leads to doubting the results of the epistemological process before investigating the bases of knowledge; i.e., principles, hypotheses, and results of hypotheses testing (Saad, 2018).

Most of the works used the theory as a theoretical approach in their studies of new media. Thus, despite its criticism, the theory is used in the local context.

All studies projected some of the theory assumptions on the local communicative context about the use of modern technology, such as testing the social change, the medium is the message, the law of media, and the use of smartphone. In addition, the values, social construct, and complexity paradigms are introductions that can explain and interpret technological determinism. Besides, we find diversity of interpretation methods, including the historical, desk, and critical. The first paper is an important theoretical and methodological reference for understating the digital McLuhanism that is still absent from the Algerian academic scene. Seemingly, the author explained and analyzed it and suggested the complexity and social construct of media paradigms to criticize the technological determinism (Al Ayadi, 2021). Moreover, the paper shows the methodological orientation towards the qualitative interpretational studies as methodological

introductions to adapt the ideas of McLuhan to the digital context.

Abdullah Thani Mohamed Nadir (2017, p. 77) sees that the epistemology of information and communication sciences depends on the development of the tool or the medium, as the epistemological efforts in this field date back to the history of the philosophy of media in a technological pattern. Therefore, it is necessary to:

3. Identify the main methodological and theoretical introductions in adopting the technological determinism in the academic studies in the context of the new media:

The qualitative studies:

They rely on interpretation, analysis, and explanation to reveal the main causes of the phenomenon and of its existence. Its main methods are:

Content analysis:

It relies on analysis, interpretation, and revelation of the phenomena. Pearson defines it as one of the scientific research methods that aim at the objective, regular, and quantitative description of the apparent content of one of the communication subjects. Besides, Paul Henry and Serge Muscovy see it as a set of overlapping techniques used when discussing the linguistic tools. This method requires a set of steps, including:

First: Identification of the objectives, problematic, and assumptions.

Second: Preparation of the analysis tools.

Third: Choice of the documents to be analyzed, in addition to the population that is a set of well-defined statistical units to get data. In addition, it is necessary to choose the study sample, which Mohamed Abd al Hamid defines as unlimited items that the researcher will methodologically deal with.

Discourse analysis method: It is an outcome of linguistics. Mangueneau sees it as a contact point between the linguistic thinking and the other human sciences. Discourse is an important speech from the perspective of the reasoning mechanism. Thus, discourse analysis is a linguistic study of the conditions of producing the text. The discourse must be audible and needs a speaker and a listener; the first has an influence on the second (Becharif, 2018).

-The ethnographic method: Marshall defines it as the process of directly observing a social group and making a written description. Thus, ethnography relies on description and observation to answer questions on the life and behaviors of a group or individuals in a given time. It needs the participation in the informants' experiences to project the scientific efforts and theories on the phenomenon.

-Ethnography tools: For data collection, ethnography uses:

a- Observation: It is one of the scientific methods for collecting data on the individuals behaviors in different conditions and situations. It is defined as the tool through which we examine the apparent behavior of people and know the surrounding conditions, behaviors, interactions, relations, and lifestyles.

The ethnographic method requires using participative observation, where the researcher is part of the population, such as living with smart phone users to know their behaviors, or with different families to find out more about their uses and exposure to different technological media. Such observation needs a prior preparation and long time. Besides, it requires exact information and data and rapid recording of notes.

b. Interview:

In addition to the observation, the researcher has to use interviews to collect information. An interview is defined as a planned meeting between the researcher and the informant to answer a set of questions about a given topic. The questions are well studied and prepared by the researcher to get deep detailed information. Thus, it is important and suitable for the ethnographic studies to find out how people perceive a problem or react towards it.

As for internet or virtual ethnography, we need the same tools, but virtually, such as electronic observation through logging in the virtual world and living with the study audience to notice their behaviors and use of multimedia. Besides, the study tool is virtual to reveal the correlations between the informants and surrounding communities.

In Algeria, in his in-depth research into audience and media studies, Ali Kessaissia's call was the initial step for adopting the ethnographic approach in media research. His advocacy for adopting the ethnographic approach is based on the fact that, despite being primarily pragmatic—that is, relying only on available sources and the targeted data of the research—it remains the most appropriate approach that allows access to the worlds and contexts of families regarding the reception of media messages, both as an individual and social act. It allows for the description of these families' actions and signals in relation to their use of new information and communication technologies. Thus, the ethnographic approach focuses on understanding behavior within a social context through the researcher's active, participatory involvement within the studied group (Maliya, 2018).

This type of study was also advocated by Marshall McLuhan in his study of media and society, where he criticized Wilbur Schramm's study on Television and the Child.

Quantitative Studies

Quantitative studies primarily rely on description rather than analysis to study phenomena. The main methods include:

The Experimental Method:

It is the type of research that uses an experiment to test a hypothesis that establishes a relationship between two factors or variables, by studying corresponding situations where all variables are controlled, except for the variable whose effect the researcher is interested in studying.

The Survey Method:

It is defined an exploratory study carried out for uncovering societal problems, or a systematic attempt to report, analyze, and interpret the current state of a social system or a specific group, focusing mainly on the present time.

The Descriptive Method:

It is the precise observation and monitoring of a specific phenomenon or event in a quantitative or qualitative manner over a certain period of time, or several periods, in order to identify the phenomenon or event in terms of content,

and to reach results and generalizations that help in understanding and developing reality.

The descriptive and survey methods rely on the questionnaire to reach results that explain the phenomenon under study. A questionnaire is a set of written questions prepared for obtaining information or opinions from respondents about a specific phenomenon or situation, and is one of the most widely used tools for collecting data in the social sciences (Bacherif, 2018).

Theoretical Research Approaches

The theoretical research approaches that enable the adoption of Technological Determinism in media and communication studies involve examining the theory's foundations and its key assumptions to apply them to the study, and formulating study hypotheses that align with the theory's assumptions. Moreover, some theories rely on a specific model that explains and interprets their view of the phenomenon.

Adopting any theory in preparing an academic study in the field of Media and Communication Sciences obliges the researcher to take an important step: establishing an element known as the theoretical Background and its application in the study. This requires knowing the key assumptions on which the theory is based,

and then demonstrating how to benefit and apply it.

The theory in question is based on three main assumptions:

- Media shape the sensory perceptions and cognitive organization of individuals within society.
- Media redistribute perceptual patterns across different social systems.
- Every medium reshapes the symbolic and cultural environment.

This is confirmed by McLuhan's famous quote: "We shape our tools, and thereafter our tools shape us." This illustrates the structural nature of the influence of technological variables on culture. Technological revolutions sometimes lead to profound social changes within societies, and these changes cannot be resisted without understanding how they occur and are formed.

The concepts of hot and cold Media are among the concepts presented by Marshall McLuhan, in addition to the concept of the "Non-corporeal Man" (man without a body or disembodied man) in his description of the telephone user, and the idea that media are extensions of the human senses.

The theory also yielded a law known as the Tetrad of Media Effects (Tetrad), which enables researchers in Media and Communication Sciences to apply the theory to new media studies—like smartphones and their effects—as well as 5G technology and Artificial Intelligence, and the developments in media and the accompanying changes within society.

The Tetrad proceeds from a set of questions. When analyzing any medium or technology, McLuhan raises these questions:

- What does it enhance?
- What does it make less important/retrieve?
- What does it retrieve or bring back?
- What does it reverse or flip into when pushed to extremes or overused?

(McLuhan, 1988). The importance of this law lies in its ability to:

- Provide a comprehensive analytical framework for understanding the impact of any new communication medium.
- Help researchers predict social and cultural changes before they fully materialize.

- Show that technology is not neutral, but fundamentally changes values and relationships.

Conclusion:

Based on what was said, it is necessary to review the theory of technological determinism in the Algerian media studies in the light of the digital transformations. In addition, our findings show that:

- The main assumption in McLuhanism is that the medium is the message; however, it was mistakenly interpreted that McLuhan prefers the medium and excludes the importance of the message.
- The use of technological determinism in the Algerian media studies takes various aspects, such as descriptive adoption and critical use, reflecting the lack of a unified epistemological view.
- The technological determinism can be adapted to the Algerian local context.
- The Algerian literature review shows that the technological determinism is not enough to understand the complexities of the new communication environment; therefore, it is necessary to use other epistemological paradigms,

such as the social construct, the complexity, and the values paradigms.

- Repositioning the technological determinism under the modern paradigms is a good introduction to enrich the Algerian media studies and foster its ability to interpret and analyze the challenges of the digital present.
- The technological determinism is an important theoretical introduction to study the electronic media if the notions of McLuhan are understood.
- The technological determinism is a developmental theory.

The study recommends:

- Avoiding stereotypes when using the theory of technological determinism in the new media studies.
- Adapting the theory assumptions to the Algerian local context.
- Combining the qualitative and quantitative methods and tools of approach to adopt the medium paradigm.
- Paying attention when translating the works of McLuhan and the

other scientific productions to avoid falsification.

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